

Integrated Li-Po Power Management System

Component Datasheet Summary & Specification Reference Guide | Project: 5V / 12V Selectable Power Module

This reference document provides comprehensive datasheet technical specifications, absolute maximum ratings, operating conditions, and functional pin definitions for all active ICs, discrete semiconductors, passives, and connectors used in the **Integrated PCB-Based Li-Po Battery Charging & Selectable 5V/12V Boost Power Supply Module**.

1. TP4056 — 1A Standalone Linear Li-Ion Battery Charger IC

Function / Description	Complete constant-current / constant-voltage (CC/CV) linear charger for single-cell lithium-ion batteries.
Package / Footprint	ESOP-8 / SOP-8-PP (Exposed Thermal Pad)
Input Voltage Range (Vcc)	4.0V to 8.0V (Absolute Max: 10V; 4.5V - 5.5V recommended via USB)
Programmable Charge Current	Up to 1000 mA (Configured via R_PROG connected to GND)
R_PROG Formula	$I_{BAT} = (V_{PROG} / R_{PROG}) * 1200 = (1.0V / R_{PROG}) * 1200 \rightarrow R1 = 1.2k\Omega$ yields 1000 mA
Final Charge Voltage	4.2V ± 1.5%
Status Indicators	Pin 7 CHRG (Active Low LED D1), Pin 6 STDBY (Active Low LED D2)
Safety & Protection	Soft-Start, Thermal Foldback/Regulation, Undervoltage Lockout (UVLO at 3.8V)
Pin Config (ESOP-8)	1: TEMP, 2: PROG, 3: GND, 4: VCC, 5: BAT, 6: STDBY, 7: CHRG, 8: CE (Enable)

2. DW01A — Single-Cell Li-Ion/Li-Po Battery Protection IC

Function / Description	Monitors battery voltage and current to protect against overcharge, overdischarge, and overcurrent/short-circuit.
Package / Footprint	SOT-23-6
Supply Voltage (VCC to GND)	-0.3V to +10V (Operational: 2.0V to 6.0V)
Overcharge Detection (V_CU)	4.30V ± 0.05V (Triggers OD/OC pin to cut off charging)
Overcharge Release (V_CL)	4.10V ± 0.05V
Overdischarge Detection (V_DL)	2.40V ± 0.10V (Triggers DO pin to disconnect load)
Overdischarge Release (V_DR)	3.00V ± 0.10V
Overcurrent Detection Voltage	150 mV ± 30 mV (Monitored via CS pin through 1kΩ R4)
Short Circuit Protection Delay	10 μs to 50 μs
Low Operating Current	3.0 μA (Normal Operation), 0.1 μA (Power-down Mode)
Pin Config (SOT-23-6)	1: OD (Overcharge Gate), 2: CS (Current Sense), 3: OC (Overdischarge Gate), 4: TD (Test Pin), 5: VCC, 6: GND

3. FS8205A — Dual N-Channel Common-Drain Power MOSFET

Function / Description	Dual ultra-low R_DS(on) N-channel MOSFET switch controlled directly by DW01A OD and OC pins.
Package / Footprint	TSSOP-8 / SOT-23-6 equivalent
Drain-Source Voltage (V_DS)	20V Maximum
Gate-Source Voltage (V_GS)	±12V Maximum
Continuous Drain Current (I_D)	6.0 A
Pulsed Drain Current (I_DM)	25 A
Static Drain-Source On-Resistance	R_DS(ON) < 25 mΩ @ V_GS = 4.5V, I_D = 4.5A; < 37 mΩ @ V_GS = 2.5V
Pin Config (TSSOP-8)	1,8: Drain 1 (D1); 2,3: Source 1 (S1); 4,5: Source 2 (S2); 6,7: Drain 2 (D2); Gate 1 (G1), Gate 2 (G2)

4. MT3608 — 1.2MHz 2A High-Efficiency DC-DC Boost Converter

Function / Description	Step-up DC-DC switching regulator providing regulated higher output voltage (5V/12V) from low battery input.
Package / Footprint	SOT-23-6
Input Voltage Range (V_IN)	2.0V to 24V
Output Voltage Range (V_OUT)	Up to 28V (Selectable 5V / 12V in current PCB design)
Switching Frequency	1.2 MHz Fixed PWM frequency (allows small ceramic capacitors and small 22μH inductor)
Switch Peak Current Limit	2.0 A Maximum
Efficiency	Up to 97%
Feedback Reference Voltage	V_FB = 0.600V ± 2%
V_OUT Equation	V_OUT = 0.6V * (1 + R_top / R_bottom). 5V: R10(16k) + R8(2.2k); 12V: R9(42k) + R8(2.2k)
Pin Config (SOT-23-6)	1: SW (Switch Node), 2: GND, 3: FB (Feedback), 4: EN (Enable - connected to IN), 5: IN (Power Input), 6: NC

5. SS34 — Surface Mount Schottky Barrier Rectifier

Function / Description	High-efficiency freewheeling diode placed across MT3608 SW node and OUT+ rail.
Package / Footprint	SMA (DO-214AC)
Max Repetitive Reverse Voltage (V_RRM)	40 V
Max Average Forward Current (I_AV)	3.0 A
Forward Voltage Drop (V_F)	0.50 V @ I_F = 3.0 A

Peak Forward Surge Current (I_FSM)	80 A (8.3ms half sine-wave)
Operating Junction Temp	-55°C to +150°C

6. Passive Components & Hardware Connectors Specification Summary

Component	Designators	Spec / Value	Package	Key Datasheet Note / Rating
Resistor 1.2kΩ	R1	1.2kΩ ±1%	0805 SMD	1/8W; Sets TP4056 charge current to 1000 mA
Resistor 1kΩ	R2, R3, R4	1.0kΩ ±5%	0805 SMD	1/8W; D1/D2 LED current limit & DW01A CS filter
Resistor 100Ω	R5	100Ω ±5%	0805 SMD	1/8W; DW01A VCC power supply decoupling resistor
Resistor 5.1kΩ	R6, R7	5.1kΩ ±1%	0805 SMD	1/8W; USB Type-C CC1/CC2 pull-down to GND for 5V input
Resistor 2.2kΩ	R8	2.2kΩ ±1%	0805 SMD	1/8W; MT3608 Feedback divider bottom resistor
Resistor 42kΩ	R9	42kΩ ±1%	0805 SMD	1/8W; MT3608 Feedback top resistor for 12.05V output
Resistor 16kΩ	R10	16kΩ ±1%	0805 SMD	1/8W; MT3608 Feedback top resistor for 4.96V output
Capacitor 10μF	C1, C3, C4	10μF 25V X7R/X5R	0805 / 1206	Ceramic MLCC; Input/Output smoothing capacitors
Capacitor 100nF	C2	0.1μF 50V X7R	0603 SMD	Ceramic MLCC; DW01A VCC decoupling capacitor
Power Inductor	L1	22μH ±20%	7.3x7.3mm (CDRH74)	Isat >= 1.5A, DCR < 0.11Ω; Shielded power inductor
Status LEDs	D1, D2	Red / Green	0805 SMD	V_F = 1.8V - 2.1V, I_F = 3mA - 5mA
USB-C Socket	J1	Type-C 6-Pin	SMD Receptacle	Power-only USB receptacle; 3A rated VBUS/GND contacts
SPDT Switch	SW1	Slide Switch	CAS-120A1 / PCM12	0.3A @ 6V DC; Selects between R9 (12V) and R10 (5V)
Headers	J2, J3, J4, J5	1-Pin Socket	2.54mm Pitch	3A Rated Pin Header Sockets for BAT+, BAT-, OUT+, OUT-